
Tutorial series 2 (Complex number)

Exercise 1: write in the algebraic form the following expressions:

$$\frac{3+6i}{3-4i} \quad ; \quad \left(\frac{1+i}{2-i}\right)^2 + \frac{3+6i}{3-4i} \quad ; \quad \frac{2+5i}{1-i} + \frac{2-5i}{1+i} \quad ; \quad \frac{i-4}{2i-3}$$
$$\frac{1+i}{1-i} - (1+2i)(2+2i) + \frac{3-i}{1+i} \quad ; \quad 2i(i-1) + \left(\sqrt{3}+i\right)^3 + (1+i)(\overline{1+i})$$

Exercise 2: solve in $\mathbb{C}$

$$z^3 + 2(\sqrt{2}-1)z^2 + 4(1-\sqrt{2})z - 8 = 0 \quad ; \quad z^2 - (1+2i)z + i - 1 = 0$$
$$z^2 - \sqrt{3}z - i = 0 \quad ; \quad z^2 + z + 1 = 0 \quad ; \quad 2z + i = \bar{z} + 1 \quad ; \quad 2z + \bar{z} = 2 + 3i$$
$$2z + 2\bar{z} = 2 + 3$$

Exercise 3: Solve the following systems of unknowns, the complex numbers z_1 and z_2 :

$$\begin{cases} 2z_1 - z_2 = i \\ -2z_1 + 3iz_2 = -17 \end{cases} \quad ; \quad \begin{cases} 3iz_1 + iz_2 = i + 7 \\ iz_1 + 2z_2 = 11i \end{cases}$$

Exercise 4: Put the following complex numbers into exponential form:

$$1 + i\sqrt{3} \quad 9i \quad ; \quad -3 \quad ; \quad \frac{-i\sqrt{2}}{1+i} \quad \frac{(1+i\sqrt{3})^3}{(1-i)^5}$$

Exercise 5: Calculate the square roots of:

$$3 + 4i \quad ; \quad 8 - 6i \quad ; \quad 7 + 24i \quad ; \quad -1 - i$$

Exercise 6: Find the cube roots of:

$$2 - 2i \quad ; \quad 11 + 2i \quad ; \quad -8$$

Exercise 7: Determine the set of complex numbers z such that:

$$\left|\frac{z-3}{z-5}\right| = 1 \quad ; \quad \left|\frac{z-3}{z-5}\right| = \frac{\sqrt{2}}{2} \quad ; \quad |z - 1 + i\sqrt{3}| = 2\sqrt{3}$$

Exercise 8: let the following polynomial $P(z)$ defined by:

$$P(z) = z^4 - 6z^3 + 24z^2 - 18z + 63$$

- 1) Calculate $P(i\sqrt{3})$ and $P(-i\sqrt{3})$ then determine the second-degree polynomial Q with real coefficients such that for all $z \in \mathbb{C}$, $P(z) = (z^2 + 3)Q(z)$.
- 2) solve in $\mathbb{C}$ $P(z) = 0$.